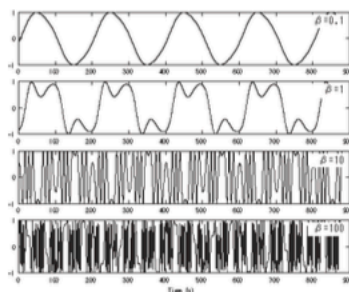


First Part: Building the dataset

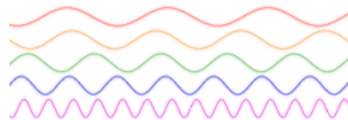
1

Implement an FM synthesizer



2

Generate a sample dataset

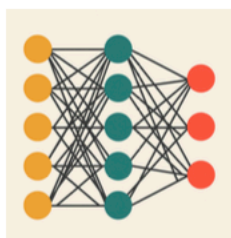


Synthesis implementation and dataset generation

Second part: Building the model

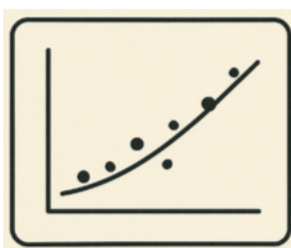
3

Guide manual experiments to design a baseline model (CNN + regressor).



4

Apply a randomized grid search to tune the model parameters.



5

Train the final model to estimate the original FM synthesis parameters

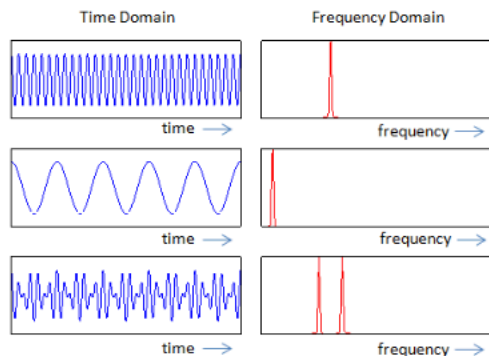


Model optimization (can be cyclic between steps 3 and 5)

Third Part: Evaluating performance

6

Estimate FM synthesis parameters from real instrument samples using the model



7

Regenerate real instrument samples



8

Measure audio similarity between the original and regenerated instrument samples

Making a performance analysis on real-world problems